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Automatic Landslide Detection

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***Abstract*— One of the most frequent natural disasters is a landslide. It is defined as the downward movement of soil and rock due to gravitational force. It leads to loss of people and property. So, landslide detection plays a crucial role in saving people's life. In this project work, we took sentinel-1 satellite dataset of Kodagu(formally known as Coorg) in Karnataka, India. The images are pre-processed using CLAHE(Contrast Limited Adaptive Histogram Equalization) and pixels based classification will be applied to extract the low-level features. The image background is removed by Digital Elevation Model(DEM). Using morphological analysis, the small objects that distinguish landslides from another are removed by taking maxima and minima points. Landslides are detected from that image using H-BEMD(Hue—Bi-Directional Empirical Mode Decomposition). H-BEMD is the improved version of BEMD(Bi-Directional Empirical Mode Decomposition) along with Hue Channel.**

Keywords*— Contrast Limited Adaptive Histogram Equalization(CLAHE) algorithm, Digital Elevation Model(DEM), Morphological Analysis, Convolutional Neural Network(CNN), H-BEMD(Hue --- Bi-Dimensional Empirical Mode Decomposition)

I. INTRODUCTION

When a slope changes from a stable to an unstable state, landslides happen. It results in injury, death, and property damage. For many years following a slide occurrence, it also has a negative impact on several resources, including water supply, fisheries, sewage disposal systems, dams, and roads. Therefore, landslide detection is necessary for risk assessment.

Digital image processing is widely employed as a result of technological advancements. Remote sensing images captured by satellites are utilised in defence, agriculture, navigation, and other fields. Satellite image capturing techniques have vastly improved over time, and image size has correspondingly increased. It's tough to archive such photographs for future use. It takes a long time to process. Various techniques of image processing can be applied to increase the quality and storage of an image by changing some of its features.

The satellite images range substantially in terms of textural contrasts and colour changes due to the presence of some discrepancies, and they are also exceedingly intricate. As a result, using satellite data to apply processing algorithms is difficult. Furthermore, long-distance satellite data collection is hampered by unwanted interferences, decreasing image quality. This causes substantial issues in following processing stages and degrades the quality of the image. So, before applying more sophisticated image processing methods, noise-distorted satellite images need be pre-processed.

The visual effects of images are dramatically reduced when noise, such as white and black dots, is strewn across them. Noise removal using standard filtering methods is ineffective, especially for strong noise. Improving image quality and suppressing high density noise requires maintaining margins and details. Some of the most prevalent filtering techniques for salt and pepper are the Adaptive Median (AM) filter algorithm, Switching Median (SM) filter algorithm, Extreme Median (EM) filter method, Traditional Median (TM) filter algorithm and, so on. Here, we applied Contrast Limited Adaptive Histogram Equalization(CLAHE) algorithm of adaptive algorithm for pre-processing.

An analytical technique for signal processing known as empirical mode decomposition (EMD) was initially created to study nonlinear and non-stationary one-dimensional (1D) signals. A multi-resolution decomposition approach primarily used for image processing, bidimensional empirical mode decomposition (BEMD) is a two-dimensional (2D) expansion of the EMD concept. Finding local minima and maxima points, collectively referred to as local extrema points, is necessary for BEMD in order to estimate the upper and lower envelope surfaces of the data. A Convolution Neural Network(CNN) is used to identify whether landslides are present in satellite images during the deep learning phase. This study suggests a transformation technique Hue - Bi-dimensional Empirical Mode Decomposition (H-BEMD) to effectively identify landslides under various lighting situations from classed landslide images, in order to pinpoint the region and size of the landslide. Once the position of the landslide is known, we use a number of time points to calculate the size change of the slide.

II. NOISE REMOVAL TECHNIQUES

The effect of undesired signals in an image is noise. It is referred to as random variances of information in an image. Noise is caused by variations in brightness, color information or contrast in an image. Techniques for introducing noise to an image include electronic transmission, film grain, sensor heat, and radio astronomy.

A. Filtering Techniques

De-noising is the process of removing noise from an image. This aids in improving image quality and simplifying image interpretation for analysis. There are several filtering strategies available.

Median Filter-The median filter, commonly known as the edge preserving filter, is an on-linear approach for denoising. It proceeds in such a way that each pixel is reacquired by the median value of neighboring pixels. The median filter keeps as much of an image's lines and borders as possible and outliers are eliminated.

Average Filter-The mean filter smoothens images by compressing the degree of intensity differences between adjacent pixels. Each pixel in the filter is replaced with the average of its neighbors, including itself. There are a few things to consider when utilizing the mean filter. These are the details:

1) A pixel with an extraordinary value can affect all pixels average value in the neighborhood.

2) The borders may become blur while interpolating other values.

Adaptive Wiener Filter - The adaptive Wiener filter may effectively reduce blur in an image produced by linear motion. This can be done in a method that finds the optimal compromise between and refined noise and converse filtering. It is a constant assessment of the current image.

III. Image Enhancement

A. CLAHE Algorithm

In this research, the CLAHE enhancement approach was used to increase image quality in a real-time system. To improve the visibility of a hazy image, contrast limited adaptive histogram equalization (CLAHE) is used.

Algorithm 1: CLAHE

Step 1: The acquisition of hazy image.
Step 2: Get a list of all the input variables used in the improvement process, including the number of regions in the column and row directions, the dynamic range, the distribution parameter type and the clip limit,.
Step 3: Preprocess these inputs by dividing the original image into several regions
Step 4: The process was applied to the tile.
Step 5: Create a clipped histogram and grey level mapping. Because each grey level has a different number of pixels, the contextual region is evenly distributed, the average number of pixels in each grey level is described.
Step 6: To make a better image, interpolate grey level mapping. Using four-pixel clusters and applying mapping, each of the mapping tiles will partially overlap in the image region, after which a single pixel will be retrieved and four mapping will be applied to that pixel. Repeat over an image to get improve pixel by interpolating between those outputs.

B. Color Conversion

The wavelengths range recorded by a sensor is known as a band, which is naturally defined by the energy wavelength. Each band of a multispectral image can be displayed separately as a greyscale image or as a blend of three bands in a color composite image. Red, green, and blue are the three basic colors of light. Computer screens may display a picture in three different bands at the same time by making each image a distinct primary color. We create a color composite image by combining these three images.

1) RGB Color model

The most popular colour model used in openCV and digital image processing is the RGB model. The colour image is composed of three channels. Every colour has a separate channel. The three primary colour components of this model are blue, red, and green. Only this proportional ratio of these three colours can create any other colour. As the value rises, the colour intensity rises; 0 stands for black.

2) HSV color model

Three channels make up the image. There are three channels: hue, saturation, and value. The primary colours are not used directly in this colour scheme. It makes use of colour to alter how people see them. When a cone is used to represent HSV(Hue, Saturation, Value) colour. Hue is a part of colour. The hue represents various colours in various angle ranges since the cone represents the HSV model. The term "saturation" refers to the percentage of a colour. This number might occasionally be anywhere between 0 and 1. One being the dominant colour and zero being the grey. The colour of grey is described as saturation. +The intensity of the selected colour is indicated by the value. Its percentage value ranges from 0 to 100. 1 is the brightest and shows the colour whereas 0 is black.

3) RGB to HSV Color Conversion

OpenCV is a good and easy method for converting the RGB image to the Hue channel image. In essence, OpenCV converts RGB to HSV channel at a rate of 8 bits per value (np.uint8 — from 0 to 255). There is a circle that displays the hue value (from 0 to 360 degrees). Full Hue values are transformed into an unsigned 16-bit integer. Image intensity, commonly known as luminance from chrominance, is separated using HSV. Weather, particularly the lighting conditions for each snap, affects satellite photographs.

IV. H-BEMD Algorithm

The BEMD algorithm has been enhanced by H-BEMD. A Grey image from RGB is subjected to Original BEMD [5], which has a value range (in 8 bit/value) of 0 to 255. The major significant key points in BEMD are recognizing extrema of an image signal, hence when applied to the Hue channel, BEMD does not produce rectified results (landslide region). As a result, the Hue value is adjusted using the sin and cosin values. The extrema of sine and cosine are then discovered. Extrema sin and cosine are finally combined using arctan2. It is the Hue channel's extreme value. The symbols θ and φ, respectively, stand for the values of sin(hue) and cos(hue). Through the H-BEMD sifting process, the sine and cosine of the color images are decomposed adaptively using H-BEMD.

Algorithm 2: H-BEMD Algorithm

Step 1: Using Morphological Algorithm, find the extrema(both maxima and minima) points.
Step 2: To create the new 2D 'envelope', join the maxima and minima points of using Radial Basis Functions (RBFs).
Step 3: Make the 2D "envelope" normal.
Step 4: Average the two envelopes to find the local mean $m\theta$.
Step 5: Go through step 1 to 4 on φ, $\omega i=\varphi - m\varphi$
Step 6: Repeat the process with $\theta=\vartheta i$ and $\varphi=\omega i$ and $i:=i+1$

V. PROPOSED SYSTEM

The system consists of two modules image enhancement and landslide detection. Figure 1 depicts the total system design. The raw image is passed through the filters to remove the noise. Then the obtained image is made enhanced by using CLAHE algorithm. RGB(RedGreen-Blue) image is converted to the HSV(Hue, Saturation, Value). Digital Elevation Model(DEM) is applied to remove unwanted objects in the image. Morphological Analysis is used to remove small objects. Then the landslide detection is used to identify landslides in an image.

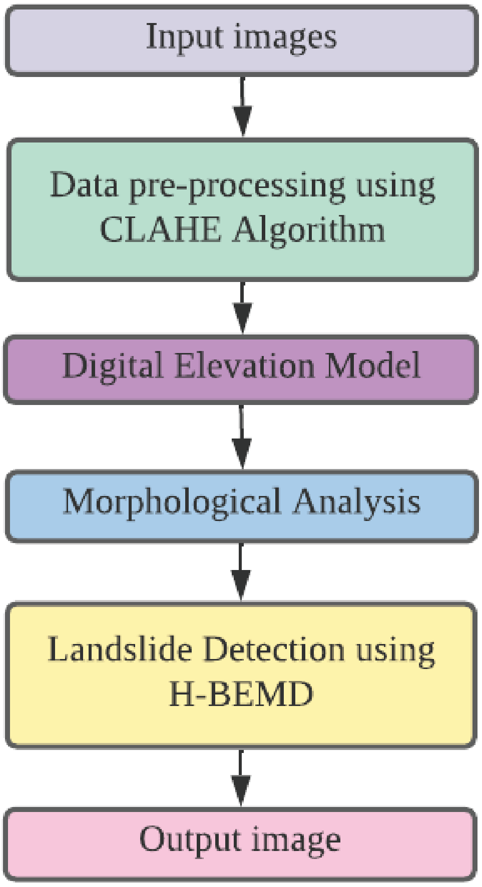

Fig 1.Proposed System

VI. RESULTS AND ANALYSIS

Using various datasets, we have implemented the landslide detection system. Here, the input image taken is Figure 2 which is in the "tiff" format. RGB input image is converted into HSV image to identify the detailings of thee image. HUE channel is used for the image conversion process. For image pre-processing CLAHE algorithm is used. Digital Elevation Model(DEM). Using morphological analysis, extreme maxima and minima points are shown in the Figure 4 of digital elevation model and the small objects that distinguish landslides from another are removed. These extreme points are averaged out, and the result is deducted from the image. The landslide features present in the image are highlighted as shown in the Figure 5 using HBEMD algorithm. The detected landslides are represented as shown in the Figure 6. The results obtained are accurate.

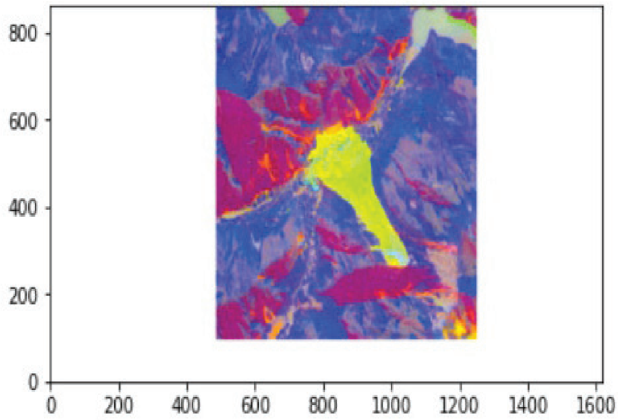

Fig 2. Input Image

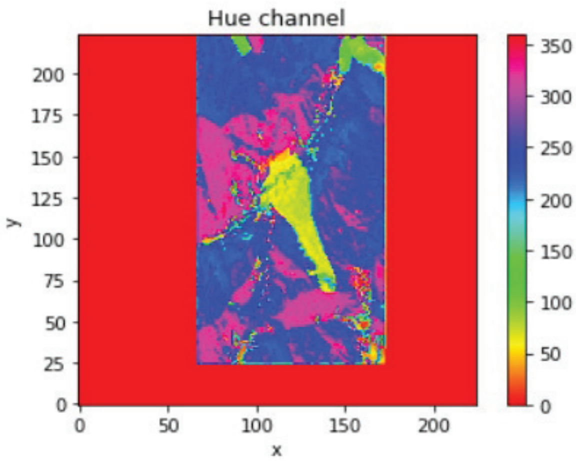

Fig 3. Hue Channel

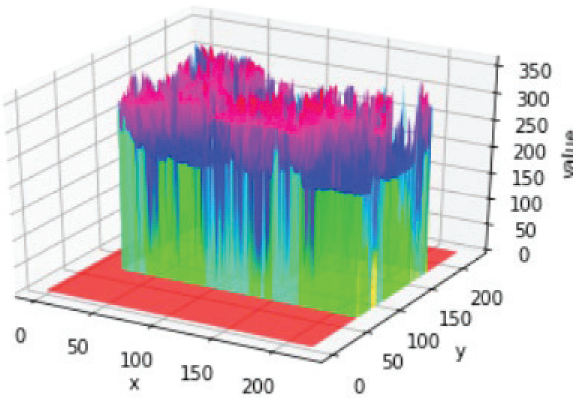

Fig 4. Maxima Minima graph

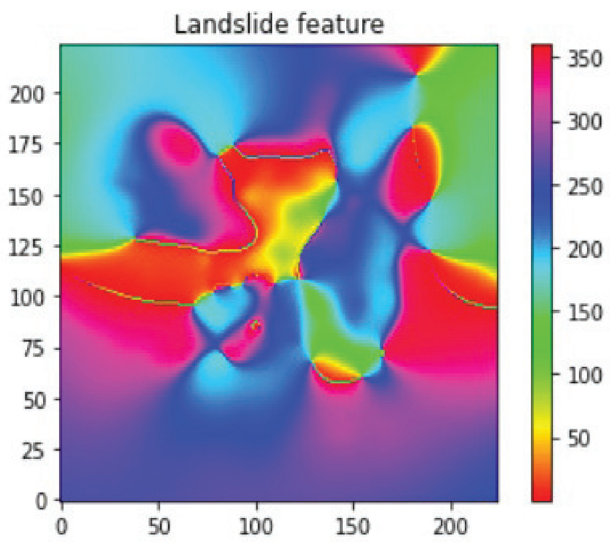

Fig 5. HSV representation of Landslides

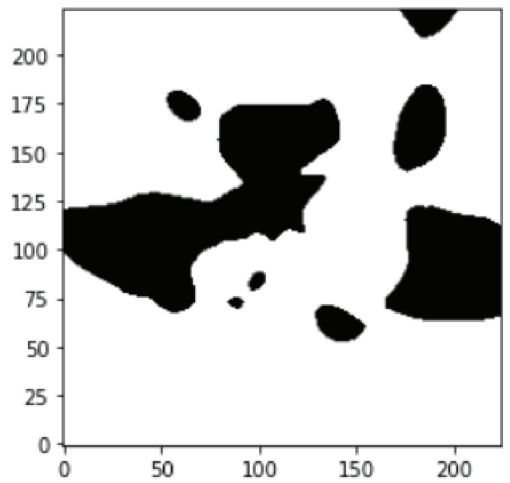

Fig 6. Output image

VII. CONCLUSION

In this study, a model is developed that detects the landslide in the sentinel-1 satellite image. BEMD(Bi-Directional Empirical Mode Decomposition) is the algorithm of Convolutional Neural Network(CNN) technique. H-BEMD(Hue--- Bi-Directional Empirical Mode Decomposition), which is the extension of BEMD(Bi-Directional Mode Decomposition) is used to identify landslides. The identified landslides are highlighted. This model can also be used for other natural disasters based on the clear identification of tectonic plate movements. From our observation, we observed that H-BEMD(Hue--- Bi-Directional Empirical Mode Decomposition) algorithm gives more accuracy than other algorithms for our project. In this work, we detected landslides. In future work, we will identify the movement of landslide. Based on the degree of the landslide movement, we can know the rate of danger which will going to occur. We can intimate to the people in that specified area to vacate and go to safe place. By using this algorithm, landslides are monitored frequently for analysis of landslide movement. With this H-BEMD(Hue--- Bi-Directional Empirical Mode Decomposition), other natural disasters can also be detected to save lives of people. This algorithm can be used for other works which needs detection of objects in satellite images.

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